Lecture 14: Causal Inference

From "What Happened?" to "What Would Happen If. . . ?"

Previously, on Lecture 13. . .

GLMs: Random component + systematic component + link function.

OLS as GLM: Normal + Identity. **Logistic:** Bernoulli + Logit. **Poisson:** Poisson + Log.

IRLS: Iteratively reweighted least squares for MLE in GLMs.

Deviance & AIC: Model comparison tools that generalize RSS and R^2_{adj}.

Today: All of L1–L13 answered "*what happened?*" and "*is the pattern real?*"

Now the hardest question: **"Did X cause Y?"**

Outline

Correlation $\neq$ Causation

Randomized Experiments

Potential Outcomes Framework

Causal Graphs (DAGs)

Observational Studies

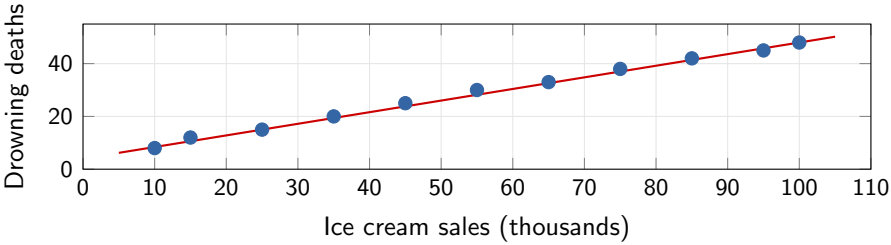
Doing vs Seeing

Practical Causal Analysis

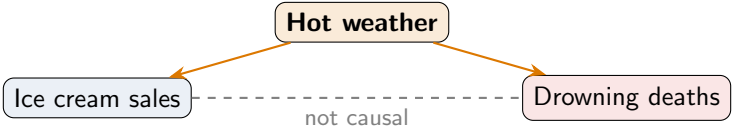
Part I: Correlation $\neq$ Causation

The most important sentence in statistics

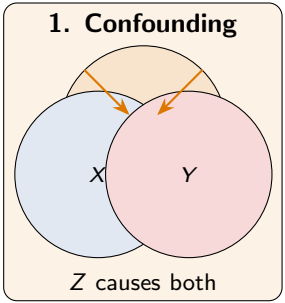
Spurious Correlations

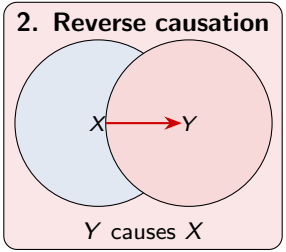


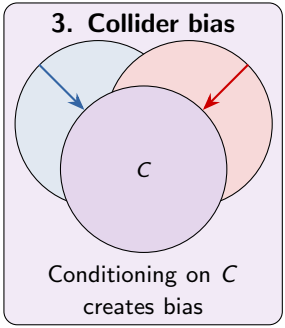
$r = 0.95$, $p < 0.001$. Banning ice cream saves lives? **No.** A **confounder** (hot weather) drives both.



Three Reasons Correlation $\neq$ Causation





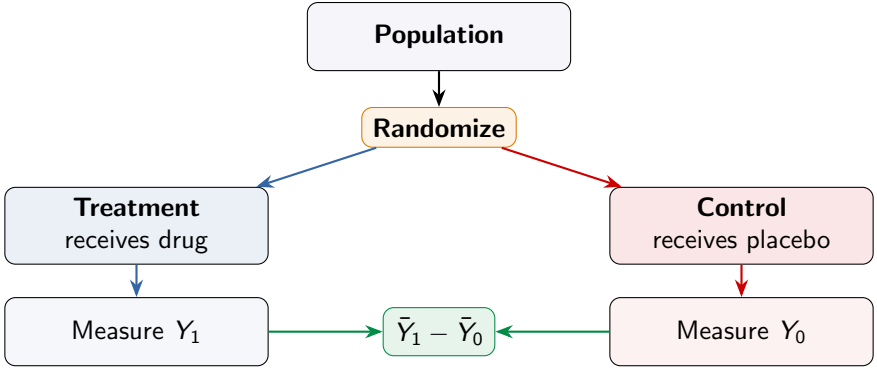


Regression, correlation, p-values — none can distinguish these scenarios. We need **additional assumptions** or **experimental design** to claim causation.

Part II: The Gold Standard

Randomized controlled trials (RCTs)

Why Randomization Works



Randomization makes groups **comparable on average** — both observed and *unobserved* confounders. Any difference in outcomes is attributable to the treatment.

From A/B Tests to RCTs

Connection to L11

	A/B Test (L11)	**RCT** (clinical trial)
Setting	Website / app	Hospital / field
Units	Users / sessions	Patients / subjects
Treatment	New button color	New drug
Control	Old design	Placebo
Randomization	Cookie / user ID	Allocation sequence
Blinding	Users unaware	Double-blind ideal
Analysis	z-test / χ^2	t-test / survival analysis

A/B testing *is* a randomized experiment.
The statistical logic is identical — only the domain differs.

Part III: Potential Outcomes

The Rubin causal model

The Fundamental Problem of Causal Inference

Patient	$Y_i(1)$ (drug)	$Y_i(0)$ (placebo)	**Causal effect** $Y_i(1) - Y_i(0)$
Alice	120	?	?
Bob	?	140	?
Carol	115	?	?
David	?	135	?

The fundamental problem: Each person is either treated or not — never both.

We can *never* observe the individual causal effect $Y_i(1) - Y_i(0)$.

The missing entries ("?") are called **counterfactuals**: what *would have happened*.

Notation: $Y_i(1)$ = outcome under treatment, $Y_i(0)$ = outcome under control. Only one is observed.

Average Treatment Effect (ATE)

$$\text{ATE} = E[Y(1) - Y(0)] = E[Y(1)] - E[Y(0)]$$

We can't observe individual effects, but we *can* estimate the **average** effect.

Under randomization:

$E[Y(1) \mid T=1] = E[Y(1)]$
$E[Y(0) \mid T=0] = E[Y(0)]$

Treatment is independent of potential outcomes.

So we can estimate ATE:

$\widehat{\text{ATE}} = \bar{Y}_{\text{treated}} - \bar{Y}_{\text{control}}$

The simple difference in means is **unbiased** for the ATE.

Other estimands: ATT (on the treated), CATE (conditional), LATE (local) — beyond our scope.

Why Randomization Identifies ATE

Step 1: We want $E[Y(1)] - E[Y(0)]$, but can only observe $Y(1)$ for treated and $Y(0)$ for control.

Step 2: Naïve comparison: $E[Y \mid T=1] - E[Y \mid T=0]$.

Step 3: This equals ATE + **selection bias**: $\underbrace{E[Y(0) \mid T=1] - E[Y(0) \mid T=0]}_{\text{selection bias}}$.

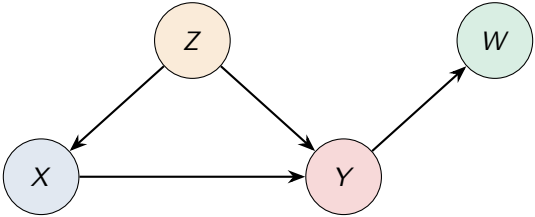
Step 4: Randomization ensures $T \perp\!\!\!\perp (Y(1), Y(0))$, so selection bias $= 0$.

Without randomization: selection bias $\neq 0$. Healthier patients may choose the new drug. Richer users may see the new feature.

Part IV: Causal Graphs

DAGs: seeing the causal structure

Directed Acyclic Graphs (DAGs)



What is a DAG?

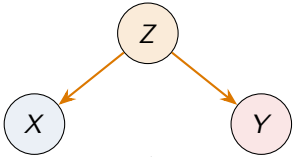
- Nodes = variables
- Arrows = direct causal effects
- "Directed": arrows have direction
- "Acyclic": no cycles ($X \to Y \to X$)

Reading the DAG:

- Z causes both X and Y (confounder)
- X also directly causes Y
- W is a consequence of Y
- No arrow = no direct effect

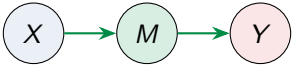
Three Fundamental Structures

Fork (Confounder)



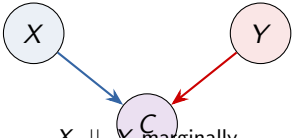
$X \not\perp\!\!\!\perp Y$
Control Z to block

Chain (Mediator)



$X \not\perp\!\!\!\perp Y$
Control M to block
(but lose indirect effect)

Collider

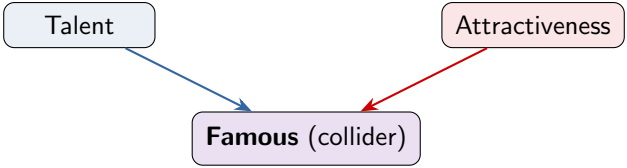


$X \perp\!\!\!\perp Y$ marginally
Control C
opens path!
(creates bias)

Key insight: Conditioning on a confounder *removes* bias. Conditioning on a collider *creates* bias.
You must know the DAG to decide what to control for.

Collider Bias: A Concrete Example

The "talent vs attractiveness" paradox



In the general population:

Talent and attractiveness are independent. No correlation.

Among celebrities only:

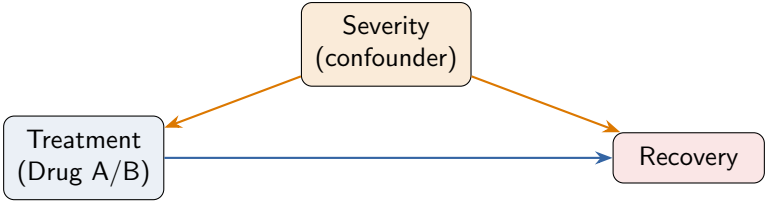
Negative correlation! Very talented celebrities tend to be less attractive, and vice versa.

By conditioning on the collider ("among the famous"), we *created* a spurious association.

Same logic: "Why are all my dates either boring or unstable?" — you're conditioning on "agreed to date me."

Simpson's Paradox Revisited

From L2 — now we have the language to explain it



	Drug A	**Drug B**	**Who's better?**
Overall	40%	50%	B
Mild	60%	55%	A
Severe	20%	15%	A

DAG tells us: Severity is a confounder ⇒ **condition on it** (stratify). Stratified result (A wins) is correct. Overall is misleading: A got more severe cases.

Part V: When You Can't Randomize

Observational causal inference

The Problem with Observational Data

RCT ✓

Randomization balances
all confounders
(observed + unobserved)

$$T \perp\!\!\!\perp (Y(1), Y(0))$$

Gold standard

Observational ✗

People self-select
into treatment

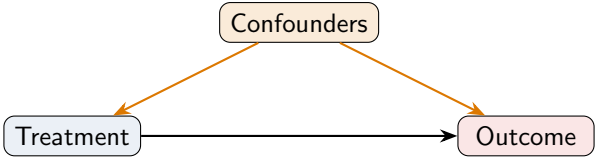
$$T \not\perp\!\!\!\perp (Y(1), Y(0))$$

Need extra work

When can't we randomize? Ethics (can't assign smoking), cost, logistics, already-collected data.
We need methods that approximate what randomization does.

Strategy 1: Conditioning / Regression

Control for confounders directly



$$Y_i = \beta_0 + \beta_1 T_i + \beta_2 Z_{i1} + \beta_3 Z_{i2} + \cdots + \varepsilon_i$$

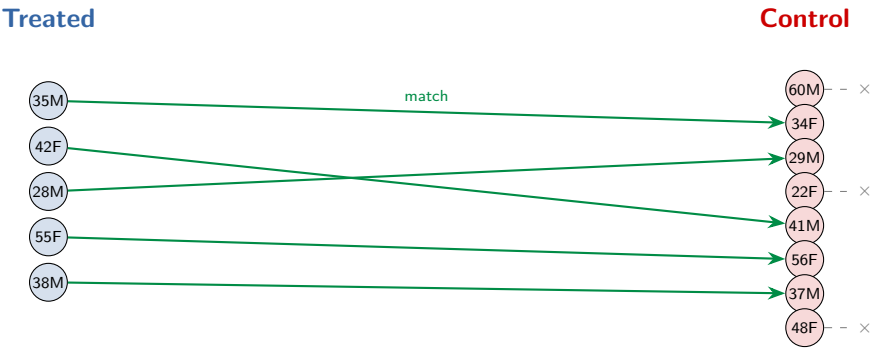
$\hat{\beta}_1$ estimates the treatment effect **after adjusting for** $Z_1, Z_2, \ldots$

Pros:
- Simple (just add covariates)
- Uses familiar tools (OLS, GLM)
- Can estimate heterogeneous effects

Cons:
- Must know *all* confounders
- Assumes correct functional form
- Can't handle unmeasured confounders

Strategy 2: Matching

Find "twins" — one treated, one not



For each treated unit, find a control with **similar covariates** (age, sex, . . .). Compare outcomes within matched pairs. Unmatched controls are discarded.

Strategy 3: Propensity Score

Rosenbaum & Rubin (1983)

$$e(X) = P(T = 1 \mid X)$$

The **propensity score** = probability of treatment given covariates.

Estimated by logistic regression: $\text{logit}(e) = X\beta$ (from L12/L13!)

PS Matching

Match on $e(X)$ instead of all X's. Solves dimensionality.

PS Weighting

Weight by $1/e$ (treated) or $1/(1-e)$ (control). IPW estimator.

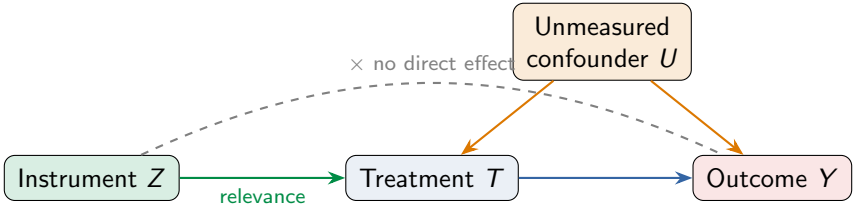
PS Stratification

Group by $e(X)$ quintiles. Estimate ATE within each, average.

Key theorem: If treatment is ignorable given X, it's also ignorable given $e(X)$ alone.

Strategy 4: Instrumental Variables

When you can't measure all confounders



An **instrument** Z must satisfy:

1. **Relevance:** Z affects treatment ($Z \to T$).
2. **Exclusion:** Z affects Y *only through* T (no direct $Z \to Y$ path).
3. **Independence:** Z is independent of unmeasured confounders.

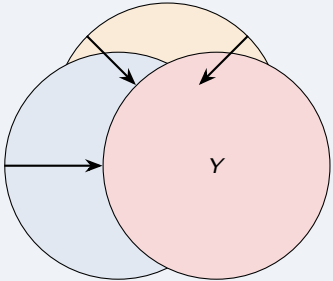
Example: Effect of military service on earnings. Instrument: draft lottery number (random, affects service, no direct effect on earnings).

Part VI: Doing $\neq$ Seeing

$P(Y \mid X)$ vs $P(Y \mid \mathrm{do}(X))$

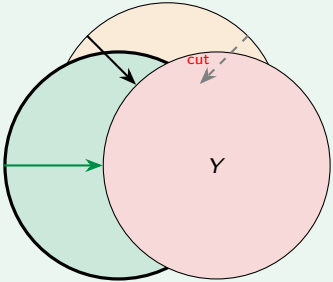
Observing vs Intervening

Observing: $P(Y \mid X = x)$



"What is Y among those who *happened to* take $X = x$?"

Intervening: $P(Y \mid \mathbf{do}(X = x))$



"If we *force* $X = x$, what is Y?" Arrows into X cut.

The do(·) operator (Pearl) formalizes intervention: set X externally, severing all causes of X.

The Backdoor Criterion

When can we estimate causal effects from observational data?

A set of variables Z satisfies the **backdoor criterion** relative to (X, Y) if:

1. Z blocks every "backdoor path" from X to Y (paths with an arrow *into* X).

2. Z contains no descendant of X.

$$P(Y \mid \mathrm{do}(X = x)) = \sum_z P(Y \mid X = x, Z = z)\, P(Z = z)$$

Adjustment formula: If Z satisfies the backdoor criterion, the causal effect is computable from observational data by conditioning on Z.

This is exactly what regression does when you "control for" confounders. The DAG tells you *which* variables to control for.

Part VII: In Practice

A checklist for causal claims

The Ladder of Causation

Pearl's three levels

Level 3: Counterfactual (Imagining)
"Would this patient have survived *if* they had taken Drug A instead of B?"
Tools: structural equation models, potential outcomes, twin networks

Level 2: Intervention (Doing)
"What *would happen* if we give everyone Drug A?" $P(Y \mid \mathrm{do}(X))$
Tools: RCTs, adjustment formula, instrumental variables, propensity scores

Level 1: Association (Seeing)
"Among people who took Drug A, what fraction recovered?" $P(Y \mid X)$
Tools: regression, correlation, hypothesis tests — everything from L1–L13

increasing difficulty

Causal Inference Checklist

1. **State the causal question** clearly. What is the treatment? What is the outcome?
2. **Draw the DAG.** Identify confounders, mediators, colliders. Use domain knowledge.
3. **Can you randomize?** If yes $\rightarrow$ RCT / A/B test. You're done (L11).
4. **If not: identify a valid adjustment set** using the backdoor criterion.
5. **Choose method:** regression, matching, propensity scores, IV, diff-in-diff, ...
6. **Check assumptions.** Sensitivity analysis: how would an unmeasured confounder change results?
7. **Report honestly.** State assumptions. Acknowledge limitations. Avoid overclaiming.

Common Pitfalls

✗ Controlling for everything

Adding all available variables to regression can *introduce* bias (collider, mediator). Let the DAG guide you.

✗ "Correlation = causation" with big data

More data doesn't fix confounding. $n = 10{,}000{,}000$ with selection bias is still wrong.

✗ Post-treatment conditioning

Controlling for a variable *caused by* treatment blocks part of the effect you're trying to measure.

✗ Ignoring external validity

RCT on college students $\neq$ effect on elderly patients. The ATE depends on who's in the study.

Summary: Causal Inference

Correlation $\neq$ causation because of confounders, reverse causation, and collider bias.

Potential outcomes: $Y(1), Y(0)$. Fundamental problem = we only see one. Goal: estimate ATE.

RCTs break confounding via randomization. A/B tests (L11) are RCTs.

DAGs encode causal assumptions. Three structures: fork, chain, collider. Back-door criterion.

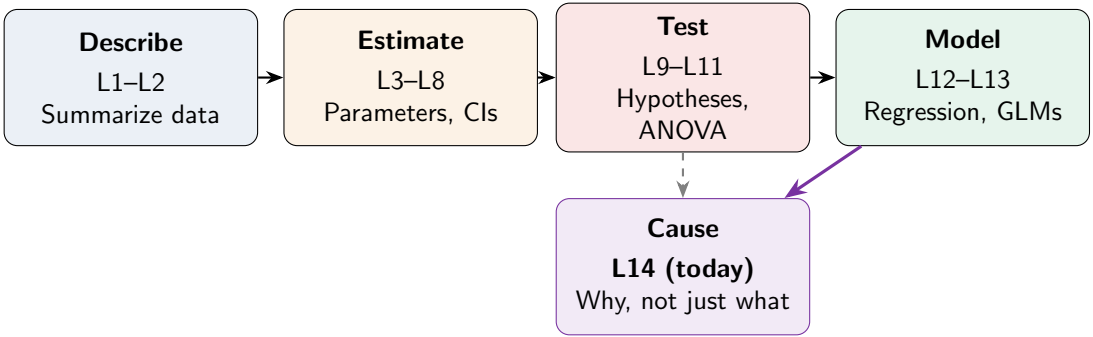
Observational methods: regression adjustment, matching, propensity scores, instruments.

do $\neq$ see: $P(Y \mid X) \neq P(Y \mid \mathrm{do}(X))$ unless we adjust for the right variables.

No free lunch: every observational method requires assumptions. State them explicitly.

The Course So Far

From data to causation



Practical Exercise

1. **Draw DAGs:** Education $\rightarrow$ Income and Exercise $\rightarrow$ Weight loss. Identify confounders and mediators in each.
2. **Simpson's paradox:** Create a $2 \times 2 \times 2$ table where overall and stratified effects reverse. Verify with a DAG.
3. **Propensity scores:** Simulate confounded data. Compare naïve ATE, PS-matched ATE, and true ATE.
4. **Collider bias:** Simulate independent talent and attractiveness. Condition on "famous" (= high talent OR attractiveness). Show spurious correlation.

Homework

1. Surgery A has higher mortality than B in hospital data. (a) Draw a DAG with severity as confounder. (b) How would propensity scores estimate the causal effect? (c) What assumptions are needed?
2. Users who see more ads buy more ($r = 0.4$, $p < 0.001$). (a) List three alternative explanations. (b) Draw DAGs for each. (c) Design an A/B test (from L11) to answer the causal question.
3. Collider simulation: $X, Y \sim N(0, 1)$ independent, $C = X + Y + \varepsilon$. (a) Compute $\text{cor}(X, Y)$. (b) Compute $\text{cor}(X, Y \mid C > 2)$. Explain.
4. "Distance to hospital" as IV for "receiving surgery." Check each IV assumption. Which is hardest to defend?

Recommended Resources

Hernán & Robins: *Causal Inference: What If* (2020) — free online. The modern standard.

Pearl, Glymour, & Jewell: *Causal Inference in Statistics: A Primer* (2016) — concise, DAG-focused.

Cunningham: *Causal Inference: The Mixtape* (2021) — free online, economics perspective.

Angrist & Pischke: *Mostly Harmless Econometrics* (2009) — IV, regression discontinuity, diff-in-diff.

Python: `dowhy` (Microsoft), `causalml` (Uber) — causal inference libraries.

Questions?

This concludes the Statistics lecture series.

L1–L14: From data description to causal inference.